

Signed Native Operator Certificates at a Fixed Shift: Pairing Defects, Star Norms, Operator Walks, and the Shorted Reassembly Boundary

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Abstract

TPC-61 identifies the exact normalized Gram of the literal missing-native map at a fixed nonzero shift, while TPC-62 shows that support degree and entrywise magnitudes alone lead only to phase-blind Schur certificates. We develop the complementary signed-operator route without selecting a minor, deleting a difficult row, or replacing a complete native output by a reduced collision cell.

Before erasure the normalized residual Gram is $I + \Xi_{=s}$, not generally the identity. Native erasure adds the new unequal-residual gate $\Xi_{\neq s}$, so the full defect is $H = \Xi_{=s} + \Xi_{\neq s} = \Xi_{\neq p}$. After the common residual Möbius gauge is calibrated, every off-diagonal entry of this full defect is exactly a normalized two-affine sum with sign $\mu(mj + h_0)\mu(nj + h_0)$. The gauge change is spectrally inert; it is not a cancellation estimate. We prove six finite-dimensional layers. First, the missing-native norm and lower frame are the extremal eigenvalues of $I + H$, where H is a Hermitian zero-diagonal signed alias operator. Second, sign-reversing partial matchings give exact pair-defect bounds for complex literal weights. Third, the maximum row ℓ^2 -star s_* satisfies

$$s_* \leq \|H\| \leq \sqrt{\Delta} s_*,$$

and a local extremizer shows why a scalar signed row sum cannot replace the star norm. Fourth, every entrywise pair majorant P gives the optimal weighted-Schur certificate $\|H\| \leq \rho(P)$. Fifth, even operator walks and traces give

$$\|H\| \leq \left(\max_m \sum_n |(H^{2q})_{mn}| \right)^{1/(2q)} \quad \text{and} \quad \|H\| \leq (\text{tr } H^{2q})^{1/(2q)}.$$

Their literal expansions retain unequal-residual and complex-phase closed-walk cancellation that an entrywise majorant discards; on a pure same-residual walk the target signs telescope to one. Sixth, these estimates form a certificate lattice for both extremal eigenvalues of the missing-native Gram.

For the full represented-plus-missing map we retain the nuisance space $\text{Ker } A$. An exact shorted-Gram corollary separates the direct canonical Gram defect from a positive shorting penalty; control of the missing-native Gram alone does not control that penalty. Five explicit countermodels distinguish coordinate gauge, pair, star, support-phase, and nuisance obstructions. All operator identities are L0 and their specialization to the complete fixed- h_0 carrier is L1. Uniform pairing defects, star savings, growing operator-walk estimates, missing-mass savings, and shorting control are stated only as unproved L2 premises. No parity, prime-pair, or twin-prime conclusion is claimed.

Keywords: fixed shift; missing-native map; signed Gram; pairing defect; star norm; Perron majorant; operator walk; shorted Gram.

Convention. The arithmetic specialization uses one fixed nonzero shift h_0 , the complete literal missing-high carrier, all missing-active physical rows, complete native output labels, and raw counting-measure atomic normalization. The shift is not specialized to 2.

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1 Introduction

The missing-high reassembly problem contains two different questions. The first asks how much a family of literal missing atoms can amplify after its residual labels are erased and its contributions are coherently returned to native outputs. The second asks whether the resulting missing map can cancel a represented map along coefficient directions invisible to the represented denominator. The first is a signed native-Gram problem. The second is a shorted-operator problem. Conflating them turns a useful restricted norm estimate into a false uniform reassembly statement.

TPC-60 defined the physical missing-high map and proved exact nuisance elimination by a shorted Gram [3]. TPC-61 then fixed the complete missing carrier before the coefficient vector, factored the map through residual coordinates, and identified its optimal normalized native Gram [4]. TPC-62 supplied inverse localization and phase-blind Schur bounds: a large operator norm leaves a row, cell, and pair witness, whereas a uniform pair estimate must still beat the actual collision degree [5]. TPC-67 pursued a different door, using all-row full-output minors, matching holonomy, and determinant mass to certify a lower frame [6]. The present paper returns to the direct operator door advertised there.

1.1 The narrow object of this paper

The main arithmetic object is only the *literal missing-native map*

$$M_X : \ell^2(\mathcal{M}_X) \longrightarrow \ell^2(\mathcal{I}_X^{\text{nat}})$$

from the TPC-61 very-high missing carrier. Its source rows, atoms, external blocks, side labels, orbit labels, residual labels, native labels, masks, profiles, and coefficients are frozen before a coefficient vector is chosen. We neither extend the TPC-61 entry formula to the entire prime packet nor assert that it already describes the cross terms of the full represented-plus-missing Gram.

There are two output levels. Before erasure the residual output is (i, s) ; after erasure the native output is i . Every still-orthogonal physical label, including side, external block, opposite row, the γ -label, and orbit time j , remains inside i . Only the residual label s is summed out, because only that erasure occurs in the physical construction. This is stricter than replacing i by a reduced arithmetic collision cell. The pre-erasure normalized Gram is not assumed to be the identity: its same-residual defect remains part of the full native operator.

1.2 The six exact layers

After physical row normalization, write

$$K = F^*F = I + H.$$

The six finite-dimensional layers proved below are:

- (A) an exact gauge-to-genuine-sign identity for H ;
- (B) complex-weight sign-reversing pair certificates;
- (C) exact star-norm bounds and their local sharpness;

- (D) an entrywise pair majorant and its Perron/weighted-Schur optimum;
- (E) even operator-walk and trace certificates that retain cancellation between complete walks; and
- (F) a certificate lattice converting any of these estimates into upper and lower eigenvalue bounds for K .

These layers are deterministic. They do not assume random signs, independence, generic phases, or an average over shifts.

1.3 Why operator walks are a genuinely different door

A pair majorant first takes an absolute value of every off-diagonal entry. Once that is done, cancellation between different row pairs is lost. By contrast, H^{2q} expands into complete row walks. Products of the row gauges telescope on closed walks, while unequal-residual target signs and literal complex phases remain. Pure same-residual target signs also telescopically trivialize, a limitation made explicit below. An arithmetic estimate for a whole decorated walk sum could therefore succeed even when every pair-majorant bound is too expensive. This is a logical possibility, not a theorem that such an estimate exists.

1.4 Claim levels and nonclaims

We use the following hierarchy throughout.

- **L0:** exact finite-dimensional linear algebra for a fixed coefficient array.
- **L1:** identification of that array with the complete literal fixed- h_0 missing carrier and its physical atomic normalization.
- **L2:** any new uniform cancellation or saving for the two-affine Möbius sums, their stars, or their operator walks.

The paper proves L0 and records the L1 crosswalk. It does not prove the L2 premises needed for a subpower native norm, an endpoint reassembly inequality, a parity improvement, a prime-pair estimate, or a twin-prime result. In particular, all statements hold for an arbitrary fixed nonzero h_0 ; none is specialized to $h_0 = 2$.

1.5 Organization

[Section 2](#) freezes the physical carrier and derives the signed Gram. [Section 3](#) gives pair certificates. [Section 4](#) proves star and Perron bounds, and [section 5](#) gives the walk route. [Section 6](#) assembles the exact certificate lattice. [Section 7](#) restores the nuisance directions of the full reassembly problem. [Section 8](#) gives five separating examples. [Sections 9](#) and [10](#) state the conditional arithmetic targets, kill criteria, and precise stopping line.

2 The complete literal missing-native interface

This section specializes the TPC–61 construction without changing its carrier. It is included to make the quantifiers and the place where signs become physical completely explicit.

2.1 Frozen carrier and complete output labels

Fix a scale X , a nonzero integer h_0 , one direct side, one external block, all source and opposite-row blocks, residue and orbit classes, masks, profiles, endpoint restrictions, and the very-high cutoff before choosing a coefficient vector. The opposite direct side is included by orthogonal direct sum. Let $\mathcal{W}_M$ be the complete set of actual nonzero missing-high atoms in this scope. Let $\mathcal{M}_X$ be the complete declared physical row set and

$$\mathcal{K}_X := \ell^2(\mathcal{M}_X) \quad (1)$$

the full coefficient space.

Every $w \in \mathcal{W}_M$ canonically recovers

$$m = m(w) = d_m \ell_m, \quad j = j(w), \quad n(w) = m(w)j(w) + h_0,$$

its largest prime $p(w) > y$, cofactor $r(w) = n(w)/p(w)$, and residual label

$$s(w) = d_{m(w)} r(w).$$

The squarefree physical support gives

$$\mu(s(w)) = \mu(d_{m(w)})\mu(r(w)).$$

The complete native key is denoted by $i(w)$. It retains the side, external block, opposite row, γ -coordinate, orbit time j , and every other label that remains orthogonal after native reassembly. Thus

$$i(w) = i(w') \implies j(w) = j(w'),$$

but the residual labels $s(w)$ and $s(w')$ may differ.

The literal atom driven by row coefficient ζ_m is

$$\beta_w \zeta_{m(w)},$$

where $\beta_w \neq 0$ retains every physical source shape, mask, multiplier, opposite-row factor, profile, and terminal weight. There is no freely chosen atomwise phase. The TPC-61 singleton incidence says that, for a fixed pair (i, m) , at most one atom of $\mathcal{W}_M$ occurs. Define

$$\mathcal{M}_M := \{m : \text{some } w \in \mathcal{W}_M \text{ has } m(w) = m\}.$$

Rows in $\mathcal{M}_X \setminus \mathcal{M}_M$ have identically zero missing mass. They remain in $\mathcal{K}_X$ and in the represented map; only the normalized missing Gram will be restricted to its active range.

2.2 Residual map, erasure, and native map

Set

$$c_w := -\mu(s(w))\beta_w, \quad (L_X \zeta)_w := c_w \zeta_{m(w)} \quad (\zeta \in \mathcal{K}_X). \quad (2)$$

The physical high-character and residual Möbius sign occur exactly once in c_w . Put

$$(T_X \zeta)_{i,s} := \sum_{\substack{w \in \mathcal{W}_M \\ i(w)=i, s(w)=s}} c_w \zeta_{m(w)}, \quad (3)$$

$$(Ez)_i := \sum_s z_{i,s}. \quad (4)$$

Then the literal missing-native map is exactly

$$\boxed{M_X = ET_X}. \quad (5)$$

This is the TPC–61 sign convention. In particular, inserting a second Möbius sign after (4) would change the physical map.

Before erasure, multiplication by

$$(U_\mu z)_{i,s} := \mu(s) z_{i,s}$$

is unitary on the residual space. This fact says only that storing the common residual sign in the input or in a signed erasure is a coordinate choice. It does *not* say that the sign disappears from EU_μ : after s is erased, different residual coordinates meet and their relative signs are genuine.

2.3 Physical atomic normalization

For every row $m \in \mathcal{M}_X$, set

$$W_M(m) := \sum_{\substack{w \in \mathcal{W}_M \\ m(w)=m}} |\beta_w|^2 \geq 0, \quad D_M := \text{diag}(W_M(m))_{m \in \mathcal{M}_X}. \quad (6)$$

This is raw counting-measure atomic mass. No output-fiber degree or collision multiplicity is divided out; equivalently,

$$D_M = L_X^* L_X$$

on the full coefficient space. Let

$$S_M := \text{Ran } D_M = (\text{Ker } D_M)^\perp = \ell^2(\mathcal{M}_M). \quad (7)$$

If $S_M = \{0\}$, then $\mathcal{W}_M = \emptyset$ and $M_X = 0$; the missing-native analysis is vacuous. Henceforth assume $S_M \neq \{0\}$.

Define the normalized missing-native incidence

$$F := M_X D_M^{\dagger/2}|_{S_M} : S_M \longrightarrow \ell^2(\mathcal{I}_X^{\text{nat}}). \quad (8)$$

Thus zero missing rows remain in $\mathcal{K}_X$, while the normalized Gram acts only on the support of the physical missing diagonal.

Define also the normalized residual incidence

$$F_{\text{res}} := T_X D_M^{\dagger/2}|_{S_M}. \quad (9)$$

The TPC–61 residual/native split gives Hermitian zero-diagonal operators

$$H_{\text{res}} := \Xi_{=s}, \quad H_{\text{erase}} := \Xi_{\neq s}, \quad (10)$$

such that

$$F_{\text{res}}^* F_{\text{res}} = I_{S_M} + H_{\text{res}}, \quad (11)$$

$$F^* F = I_{S_M} + H_{\text{res}} + H_{\text{erase}} = I_{S_M} + \Xi_{\neq p}. \quad (12)$$

In particular, the residual Gram in (11) is not generally I . Multiplication by U_μ leaves it unchanged, because two contributions meeting at one residual output have the same s . The operator H_{erase} consists exactly of the new cross terms created by erasing unequal residual labels.

Proposition 2.1 (Normalized signed alias Gram). *One has*

$$K := F^* F = I_{S_M} + H, \quad (13)$$

where

$$H = H_{\text{res}} + H_{\text{erase}} = \Xi_{\neq p} \quad (14)$$

is Hermitian and has zero diagonal. Moreover

$$\kappa_M := \sup_{\substack{\zeta \in \mathcal{K}_X \\ \langle D_M \zeta, \zeta \rangle > 0}} \frac{\|M_X \zeta\|^2}{\langle D_M \zeta, \zeta \rangle} = \lambda_{\max}(K) = 1 + \lambda_{\max}(H), \quad (15)$$

$$\alpha_M := \inf_{\substack{\zeta \in \mathcal{K}_X \\ \langle D_M \zeta, \zeta \rangle > 0}} \frac{\|M_X \zeta\|^2}{\langle D_M \zeta, \zeta \rangle} = \lambda_{\min}(K) = 1 + \lambda_{\min}(H) \geq 0. \quad (16)$$

Proof. For a fixed $m \in \mathcal{M}_M$, the (i, m) -singleton property makes distinct missing atoms occupy distinct native coordinates within that column. Hence

$$\|Fe_m\|^2 = \frac{1}{W_M(m)} \sum_{\substack{w \in \mathcal{W}_M \\ m(w)=m}} |\beta_w|^2 = 1.$$

Thus $K = F^*F$ is positive semidefinite with unit diagonal. Therefore $H = K - I$ is Hermitian with zero diagonal. Substituting $x = D_M^{1/2}\zeta \in S_M$ into the two Rayleigh quotients gives the extremal eigenvalue identities. The map M_X vanishes on $\text{Ker } D_M$, so retaining zero missing rows in $\mathcal{K}_X$ does not change either quotient. $\square$

Remark 2.2 (Notation). The symbol α_M in (16) is the lower eigenvalue of the atomic-normalized missing-native Gram. It is not the restricted erasure upper norm denoted by β_M in TPC-61.

2.4 Exact removal of the row gauge

Define the diagonal row gauge

$$U_d := \text{diag}(\mu(d_m))_{m \in \mathcal{M}_M}. \quad (17)$$

It is a self-adjoint unitary on the squarefree active support.

Theorem 2.3 (Gauge-to-genuine-sign identity). *Put $\tilde{H} := U_d H U_d$. Then $\tilde{H}$ is unitarily equivalent to H , has the same zero diagonal, spectrum, entry magnitudes, star norms, and collision graph, and for $m \neq n$,*

$$\tilde{H}_{mn} = \frac{1}{\sqrt{W_M(m)W_M(n)}} \sum_{\substack{w, w' \in \mathcal{W}_M \\ m(w)=m, m(w')=n \\ i(w)=i(w')}} \mu(mj(w) + h_0) \mu(nj(w') + h_0) \overline{\beta_w} \beta_{w'}. \quad (18)$$

Every label summed in (18) is an actual complete native output; no reduced collision cell has been substituted.

Proof. The normalized matrix entry is

$$K_{mn} = \frac{1}{\sqrt{W_M(m)W_M(n)}} \sum_{\substack{w, w' \in \mathcal{W}_M \\ m(w)=m, m(w')=n \\ i(w)=i(w')}} \mu(s(w)) \mu(s(w')) \overline{\beta_w} \beta_{w'}.$$

The two minus signs in the definition of M_X cancel. For $m \neq n$, this is H_{mn} . Multiplication by the row gauges replaces $\mu(s(w))$ by

$$\mu(d_m) \mu(s(w)) = \mu(r(w)).$$

Since $mj(w) + h_0 = p(w)r(w)$ is squarefree and $p(w) \nmid r(w)$,

$$\mu(mj(w) + h_0) = -\mu(r(w)).$$

The two extra minus signs again cancel in the product for m, n , giving (18). Unitary conjugation proves every spectral and magnitude assertion. $\square$

2.5 Direct-cell formula and the orthogonal second side

Assume in this subsection the identical-support direct-cell specialization of TPC-61: after the literal mask and opposite-row factors are retained, the high and missing tests depend on (m, j) and not on the surviving γ -coordinate. On the displayed left direct cell, write the unique atom as $w = (m, j, \gamma)$, extending its coefficient by zero when the declared literal support fails, and factor

$$\beta_{m,j,\gamma} = a_m(j)K_{m\gamma}q_\gamma(j). \quad (19)$$

Here $a_m(j)$ contains the source and terminal shapes, $q_\gamma(j)$ contains the opposite-row shape and its fixed output phase, and $K_{m\gamma} \in \{0, 1\}$ is the literal static mask. Define the actual missing-time indicator

$$\mathbf{m}_m(j) := \mathbf{1}_{\{(m,j,\gamma) \in \mathcal{W}_M \text{ for some } \gamma\}}, \quad (20)$$

where coefficient zeros caused only by $K_{m\gamma}$ or $q_\gamma(j)$ remain inside those factors. Put

$$\Gamma_{m,n}(j) := \sum_{\gamma} K_{m\gamma} K_{n\gamma} |q_\gamma(j)|^2. \quad (21)$$

Corollary 2.4 (Literal direct-cell two-affine formula). *For distinct active rows m, n , the left direct-cell contribution is*

$$\boxed{\begin{aligned} \tilde{H}_{mn}^L &= \frac{1}{\sqrt{W_M(m)W_M(n)}} \sum_j \mathbf{m}_m(j) \mathbf{m}_n(j) \\ &\quad \times \mu(mj + h_0) \mu(nj + h_0) \overline{a_m(j)} a_n(j) \Gamma_{m,n}(j). \end{aligned}} \quad (22)$$

The right direct slice has the symmetric formula with the two row systems exchanged. Its output range is orthogonal to the left slice, so

$$K = F_L^* F_L + F_R^* F_R.$$

Consequently, for $m \neq n$,

$$H_{mn} = H_{mn}^L + H_{mn}^R,$$

with no cross-slice Gram term.

Proof. A common complete native output on the left fixes the same pair (γ, j) . Substitute (19) into (18), retain the two actual missing indicators, and sum over γ . Orthogonality of the declared side labels gives the final assertion. $\square$

Outside the declared identical-support specialization, the compressed $\Gamma_{m,n}(j)$ -formula need not be valid. One must either retain the general literal pair sum (18), or replace $\mathbf{m}_m(j)$ by the full γ -dependent first-failure indicator inside the γ -sum. No later theorem requires the compressed formula.

Corollary 2.5 (Full versus erasure-only certificates). *Every certificate developed below applies directly to the full defect*

$$H = H_{\text{res}} + H_{\text{erase}}.$$

The row gauge U_d conjugates both summands separately, so the same decomposition holds in the gauged coordinates used below. Alternatively, it may be applied separately to the two summands, followed by

$$\|H\| \leq \|H_{\text{res}}\| + \|H_{\text{erase}}\|. \quad (23)$$

A certificate for H_{erase} alone does not bound κ_M unless H_{res} is controlled separately.

Proof. The first statement is definitional, and the second is the operator triangle inequality combined with (14). $\square$

Proposition 2.6 (Same-residual target signs are frozen). *Fix distinct rows m, n . If a common-output pair (w, w') also satisfies*

$$s(w) = s(w') =: s,$$

then its gauged target sign is

$$\boxed{\mu(mj(w) + h_0)\mu(nj(w') + h_0) = \mu(d_m)\mu(d_n)}, \quad (24)$$

which is independent of the collision pair. Thus H_{res} has no intra-pair oscillation coming from the two target Möbius signs. Any cancellation there must come from the literal complex weights; new residual-sign interference occurs in H_{erase} , or in cancellation between the two completed operators.

Proof. Squarefreeness gives

$$\mu(s) = \mu(d_m)\mu(r(w)) = \mu(d_n)\mu(r(w')).$$

Since each row gauge is a sign,

$$\mu(mj(w) + h_0) = -\mu(r(w)) = -\mu(d_m)\mu(s),$$

and similarly for n, w' . Multiplying the two identities proves (24). Before the row gauge, the two residual signs at the same (i, s) multiply to $\mu(s)^2 = 1$. $\square$

Remark 2.7 (What the identity does not prove). The theorem exposes the exact signs on which new arithmetic would have to act. It supplies no saving for their sum. The output gauge U_μ and the source-row gauge U_d act on different spaces and must not be identified. Either diagonal gauge can reveal a useful formula, but neither can reduce an operator norm.

2.6 Six-item L1 audit

The literal specialization satisfies the following audit.

- (L1.1) The entries are the actual coefficients $-\mu(s(w))\beta_w$, with no free atomwise phase.
- (L1.2) Every target is $mj + h_0$ for one fixed physical $h_0 \neq 0$.
- (L1.3) The full diagonal D_M is raw physical atomic mass; its zero rows are retained in $\mathcal{K}_X$.
- (L1.4) Source rows and every retained output label are canonical; only the residual label is erased.
- (L1.5) The carrier, active rows, complete native outputs, collision graph, and every certificate family below are fixed before the later coefficient vector or extremizer.
- (L1.6) The exact identities cost no endpoint exponent. Every eventual arithmetic estimate must still fit the total strict $1/400$ endpoint ledger.

3 Literal pair cancellation after gauge removal

Because H and $\tilde{H}$ are unitarily equivalent, we work with the gauged matrix and drop the tilde in this and the next three sections. Thus every off-diagonal entry has the exact form (18).

3.1 Signed complex pair lists

Fix distinct active rows m, n . Let

$$\mathcal{C}_{mn} := \{(w, w') \in \mathcal{W}_M^2 : m(w) = m, m(w') = n, i(w) = i(w')\}$$

be their complete native collision list. For $\xi = (w, w') \in \mathcal{C}_{mn}$, define

$$\varepsilon_\xi^{mn} := \mu(mj(w) + h_0)\mu(nj(w') + h_0) \in \{-1, 1\}, \quad (25)$$

$$v_\xi^{mn} := \frac{\overline{\beta_w}\beta_{w'}}{\sqrt{W_M(m)W_M(n)}} \in \mathbb{C}. \quad (26)$$

Then

$$H_{mn} = \sum_{\xi \in \mathcal{C}_{mn}} \varepsilon_\xi^{mn} v_\xi^{mn}. \quad (27)$$

The coefficient v_ξ^{mn} is complex and literal. Replacing it by its magnitude before pairing would discard physical phase information.

Write

$$\mathcal{C}_{mn}^\pm := \{\xi \in \mathcal{C}_{mn} : \varepsilon_\xi^{mn} = \pm 1\}.$$

Definition 3.1 (Admissible sign-reversing matching). An admissible matching for the row pair (m, n) is a bijection

$$\tau : A \longrightarrow B$$

between subsets $A \subseteq \mathcal{C}_{mn}^+$ and $B \subseteq \mathcal{C}_{mn}^-$. The matching may depend on the fixed literal carrier, the row pair, and all literal coefficients v_ξ^{mn} . It may not depend on a coefficient vector subsequently inserted into the row space or on an extremizing eigenvector of H .

Theorem 3.2 (Complex sign-reversing pair certificate). *For every admissible matching $\tau : A \rightarrow B$,*

$$|H_{mn}| \leq \sum_{\xi \in A} |v_\xi^{mn} - v_{\tau(\xi)}^{mn}| + \sum_{\xi \in \mathcal{C}_{mn} \setminus (A \cup B)} |v_\xi^{mn}|. \quad (28)$$

Consequently the literal pair defect

$$P_{mn} := \min_{\tau \text{ admissible}} \left\{ \sum_{\xi \in \text{dom } \tau} |v_\xi^{mn} - v_{\tau(\xi)}^{mn}| + \sum_{\xi \text{ unmatched}} |v_\xi^{mn}| \right\} \quad (29)$$

is well defined, symmetric after setting $P_{nm} = P_{mn}$, satisfies

$$|H_{mn}| \leq P_{mn}, \quad (30)$$

and is computable from the finite literal coefficient array. If $P_{mn} = 0$, then $H_{mn} = 0$.

Proof. Every matched positive-negative pair contributes

$$(+1)v_\xi^{mn} + (-1)v_{\tau(\xi)}^{mn} = v_\xi^{mn} - v_{\tau(\xi)}^{mn}.$$

The remaining terms are unmatched. The triangle inequality gives (28). There are finitely many partial matchings, including the empty one, so the minimum exists. Swapping (m, n) conjugates every weight and leaves all absolute defects unchanged; hence the symmetric convention is consistent. $\square$

Remark 3.3 (Exact orthogonality is stronger than sign balance). Equal cardinalities $|\mathcal{C}_{mn}^+| = |\mathcal{C}_{mn}^-|$ do not imply $P_{mn} = 0$. Exact cancellation also requires a matching of the *complex literal weights*. Conversely, unequal cardinalities do not preclude $H_{mn} = 0$, because several complex terms may cancel without a pairwise equality. Thus P_{mn} is a sufficient certificate, not an exact formula for $|H_{mn}|$.

3.2 An ordered partial-sum certificate

Some arithmetic decompositions supply an order on the collision list rather than a literal matching. The following exact Abel certificate records what such an order would need to control.

Proposition 3.4 (Complex Abel certificate). *If $\mathcal{C}_{mn} = \emptyset$, then $H_{mn} = 0$. Otherwise let $\xi_1, \dots, \xi_N$, with $N \geq 1$, be any fixed ordering of $\mathcal{C}_{mn}$, and abbreviate*

$$\varepsilon_k := \varepsilon_{\xi_k}^{mn}, \quad v_k := v_{\xi_k}^{mn}, \quad A_t := \sum_{k \leq t} \varepsilon_k.$$

Then

$$|H_{mn}| \leq \max_{1 \leq t \leq N} |A_t| \left(|v_N| + \sum_{k=1}^{N-1} |v_{k+1} - v_k| \right). \quad (31)$$

Proof. The empty-list assertion follows from (27). For $N \geq 1$, summation by parts gives the exact identity

$$\sum_{k=1}^N \varepsilon_k v_k = A_N v_N + \sum_{k=1}^{N-1} A_k (v_k - v_{k+1}).$$

Taking absolute values proves (31), with no restriction that the complex weights lie on one ray. $\square$

Neither the partial sums A_t nor the variation of the actual physical weights are estimated here. Choosing an order after inspecting a later extremizing vector would violate the L1 quantifier; choosing it from the fixed carrier and literal pair coefficients is allowed.

3.3 Pair certificates and their limit

The matrix

$$P = (P_{mn})_{m,n \in \mathcal{M}_M}, \quad P_{mm} := 0,$$

is a symmetric nonnegative majorant of $|H|$. It contains more signed information than the raw support degree, but it has already taken an absolute value at the level of each completed row pair. Therefore it cannot see cancellation between different edges in a row star or between different complete operator walks. The next two sections distinguish these levels.

4 Star norms and the optimal pair majorant

Let

$$\mathcal{N}_H(m) := \{n \neq m : H_{mn} \neq 0\}, \quad \Delta_H := \max_m |\mathcal{N}_H(m)|$$

be the actual collision graph and its maximum degree. Define the row ℓ^2 -star

$$s_m(H) := \left(\sum_{n \neq m} |H_{mn}|^2 \right)^{1/2}, \quad s_*(H) := \max_m s_m(H). \quad (32)$$

All definitions use the completed literal entries after summing their native collision lists.

Theorem 4.1 (Exact star sandwich). *For every finite Hermitian zero-diagonal matrix H ,*

$$\boxed{s_*(H) \leq \|H\|_{2 \rightarrow 2} \leq \sqrt{\Delta_H} s_*(H)}. \quad (33)$$

Both sides are interpreted as zero when $H = 0$.

Proof. Hermitian symmetry gives

$$\|He_m\|_2 = \left(\sum_n |H_{nm}|^2 \right)^{1/2} = s_m(H),$$

so the operator norm is at least $s_*(H)$. On the other hand, Cauchy–Schwarz on every actual row gives

$$\sum_n |H_{mn}| \leq \sqrt{|\mathcal{N}_H(m)|} s_m(H) \leq \sqrt{\Delta_H} s_*(H).$$

For a Hermitian matrix, the maximum absolute column sum equals the maximum absolute row sum. Therefore

$$\|H\|_{2 \rightarrow 2} \leq \sqrt{\|H\|_1 \|H\|_\infty} = \|H\|_\infty,$$

which proves the upper bound. $\square$

Proposition 4.2 (Local phase no-go). *For every row m ,*

$$\boxed{\sup_{\substack{\|x\|_2 \leq 1 \\ \text{supp } x \subseteq \mathcal{N}_H(m)}} |(Hx)_m| = s_m(H)}. \quad (34)$$

If $s_m(H) > 0$, equality is obtained by taking

$$x_n = \frac{\overline{H_{mn}}}{s_m(H)} \quad (n \in \mathcal{N}_H(m)).$$

Proof. The upper bound is Cauchy–Schwarz. If $s_m(H) = 0$, the zero vector is admissible and gives equality. Otherwise the displayed vector has norm one and gives

$$(Hx)_m = \frac{1}{s_m(H)} \sum_{n \in \mathcal{N}_H(m)} |H_{mn}|^2 = s_m(H).$$

$\square$

Thus cancellation in one scalar row sum $\sum_n H_{mn}$, or against any other distinguished row vector, cannot prove a uniform operator estimate. The later coefficient vector is free to align with the conjugate star. A valid star route must control the ℓ^2 -mass of every actual row star.

4.1 Pair defects imply star bounds

For the pair-defect matrix P of (29), set

$$s_m(P) := \left(\sum_{n \neq m} P_{mn}^2 \right)^{1/2}, \quad s_*(P) := \max_m s_m(P),$$

and let

$$\Delta_P := \max_m |\{n \neq m : P_{mn} > 0\}|.$$

Corollary 4.3 (Pair-to-star certificate). *One has*

$$s_m(H) \leq s_m(P), \quad \|H\| \leq \sqrt{\Delta_P} s_*(P). \quad (35)$$

Proof. The first inequality follows entry by entry from (30). The support of H is contained in the support of P , so $\Delta_H \leq \Delta_P$. Apply theorem 4.1. $\square$

4.2 The Perron certificate

The star estimate still pays the worst degree by Cauchy–Schwarz. The full nonnegative pair matrix permits optimal row weights; the matrix-norm and Perron facts used below are standard [1].

Theorem 4.4 (Pair-majorant/Perron certificate). *Let P be any finite symmetric nonnegative matrix such that*

$$|H_{mn}| \leq P_{mn} \quad \text{for all } m, n.$$

Then

$$\|H\|_{2 \rightarrow 2} \leq \rho(P) = \min_{a_m > 0} \max_m \frac{\sum_n P_{mn} a_n}{a_m}. \quad (36)$$

The minimum formula is valid even when P is reducible.

Proof. For every x ,

$$|Hx| \leq P|x|$$

coordinatewise. Hence

$$\|Hx\|_2 \leq \|P|x|\|_2 \leq \|P\|_{2 \rightarrow 2} \|x\|_2.$$

Since P is real symmetric and nonnegative, $\|P\|_{2 \rightarrow 2} = \rho(P)$.

For completeness, fix $a > 0$ and put

$$c(a) := \max_m \frac{(Pa)_m}{a_m}.$$

The weighted row-sum bound gives $\rho(P) \leq c(a)$, so

$$\rho(P) \leq \inf_{a > 0} c(a).$$

Conversely, for every $\lambda > \rho(P)$,

$$a_\lambda := (\lambda I - P)^{-1} \mathbf{1} = \sum_{k \geq 0} \lambda^{-k-1} P^k \mathbf{1} > 0.$$

It satisfies $Pa_\lambda = \lambda a_\lambda - \mathbf{1} < \lambda a_\lambda$, so

$$\inf_{a > 0} c(a) \leq \lambda \quad (\lambda > \rho(P)).$$

Letting $\lambda \downarrow \rho(P)$ proves equality of the infimum with $\rho(P)$.

It remains to verify attainment. Permute P into the orthogonal direct sum of the connected components of its undirected support graph. On each nonzero irreducible block choose a strictly positive Perron eigenvector, and on every isolated zero block choose any positive entry. Concatenating these vectors gives $a > 0$ for which

$$\max_m \frac{(Pa)_m}{a_m} = \max_C \rho(P_C) = \rho(P).$$

□

Remark 4.5 (Optimality is relative to the majorant). The right side of (36) is the best diagonal weighted-Schur aggregation of the chosen pair certificates. It need not equal $\|H\|$, because replacing H by P has already discarded all phase cancellation between different row pairs.

Corollary 4.6 (Uniform pair bound plus actual degree). *If*

$$P_{mn} \leq p_* \quad (m \neq n), \quad \Delta_P \leq \Delta,$$

then

$$\|H\| \leq \rho(P) \leq \Delta p_*. \quad (37)$$

Proof. Take $a_m = 1$ in (36). $\square$

This recovers the phase-blind degree-times-pair barrier isolated in TPC-62, but the defect P_{mn} may be substantially smaller than the absolute sum of the original collision terms because opposite two-affine signs were paired first.

5 Even operator walks and closed-walk signs

Pair and Perron certificates take absolute values before different row pairs interact. Powers of the signed matrix provide a later place to take absolute values.

5.1 Power-row certificate

For an integer $q \geq 1$, define

$$\Omega_{2q}(H) := \max_m \sum_n |(H^{2q})_{mn}|. \quad (38)$$

Theorem 5.1 (Even operator-walk certificate). *For every finite Hermitian matrix H and every $q \geq 1$,*

$$\boxed{\|H\|_{2 \rightarrow 2} \leq \Omega_{2q}(H)^{1/(2q)}}. \quad (39)$$

Proof. The matrix H^{2q} is Hermitian, and its maximum absolute row and column sums agree. Therefore

$$\|H\|^{2q} = \|H^{2q}\|_{2 \rightarrow 2} \leq \sqrt{\|H^{2q}\|_1 \|H^{2q}\|_\infty} = \Omega_{2q}(H).$$

Taking $2q$ -th roots proves the result. $\square$

The case $q = 1$ already includes interference between two-edge walks:

$$(H^2)_{mn} = \sum_k H_{mk} H_{kn}.$$

It is not identical to the star estimate. In particular, $(H^2)_{mm} = s_m(H)^2$, while off-diagonal entries may cancel.

5.2 Even moments

Let $r = \dim S_M$.

Theorem 5.2 (Even-moment certificate). *For every $q \geq 1$,*

$$\boxed{\|H\| \leq (\operatorname{tr} H^{2q})^{1/(2q)} \leq r^{1/(2q)} \|H\|}. \quad (40)$$

Proof. If $\lambda_1, \dots, \lambda_r$ are the real eigenvalues of H , then

$$\operatorname{tr} H^{2q} = \sum_{\nu=1}^r |\lambda_\nu|^{2q}.$$

This sum is at least $\max_\nu |\lambda_\nu|^{2q}$ and at most r times that maximum. $\square$

For a fixed finite matrix, the middle quantity decreases to $\|H\|$ as $q \rightarrow \infty$. In an asymptotic family whose dimension grows with X , using $q = q_X \rightarrow \infty$ requires estimates uniform in both X and q_X . A fixed-moment estimate cannot silently be extrapolated to that regime.

5.3 Literal decorated-walk expansion

For $m \neq n$, abbreviate each literal summand in (27) by

$$a_{mn}(\xi) := \varepsilon_\xi^{mn} v_\xi^{mn}, \quad \xi \in \mathcal{C}_{mn}.$$

For a collision pair $\xi = (w, w')$, define

$$j(\xi) := j(w) = j(w'),$$

where equality follows from the common complete native key. Set $\mathcal{C}_{mm} = \emptyset$, consistently with $H_{mm} = 0$.

Proposition 5.3 (Exact decorated-walk identity). *For every integer $k \geq 1$,*

$$(H^k)_{mn} = \sum_{\substack{m=m_0, m_1, \dots, m_k=n \\ \xi_\ell \in \mathcal{C}_{m_{\ell-1}m_\ell}}} \prod_{\ell=1}^k a_{m_{\ell-1}m_\ell}(\xi_\ell). \quad (41)$$

Consequently

$$\text{tr } H^{2q} = \sum_{\substack{m_0=m_{2q} \\ m_0, m_1, \dots, m_{2q-1} \\ \xi_\ell \in \mathcal{C}_{m_{\ell-1}m_\ell}}} \prod_{\ell=1}^{2q} \varepsilon_{\xi_\ell}^{m_{\ell-1}m_\ell} v_{\xi_\ell}^{m_{\ell-1}m_\ell}. \quad (42)$$

Both identities sum the complete literal native collision lists on every edge.

Proof. Expand the matrix product

$$(H^k)_{mn} = \sum_{m_1, \dots, m_{k-1}} H_{mm_1} H_{m_1 m_2} \cdots H_{m_{k-1} n}$$

and then expand every entry by (27). Summing the diagonal at $k = 2q$ gives (42). $\square$

Proposition 5.4 (Row gauges telescope on closed walks). *Let $u_m \in \mathbb{C}$ with $|u_m| = 1$, let $U = \text{diag}(u_m)$, and put $\hat{H} = U^* H U$. In every closed-walk term of $\text{tr } \hat{H}^{2q}$, the product of the row-gauge factors is one. Hence*

$$\text{tr } \hat{H}^{2q} = \text{tr } H^{2q}.$$

For the real self-adjoint Möbius row gauge $U_d = U_d^ = U_d^{-1}$, the same statement applies to $U_d H U_d$.*

Proof. An edge $m_{\ell-1} \rightarrow m_\ell$ contributes $\overline{u_{m_{\ell-1}}} u_{m_\ell}$. Their product around a closed walk is

$$\prod_{\ell=1}^{2q} \overline{u_{m_{\ell-1}}} u_{m_\ell} = 1$$

because $m_{2q} = m_0$. The trace identity also follows from unitary invariance. $\square$

Proposition 5.5 (Pure same-residual walks have no target-sign gain). *Consider one decorated closed-walk term in (42). If every decorated edge (w_ℓ, w'_ℓ) is a same-residual collision, then*

$$\prod_{\ell=1}^{2q} \varepsilon_{\xi_\ell}^{m_{\ell-1}m_\ell} = 1. \quad (43)$$

Therefore pure H_{res} walks have no cancellation from the target Möbius signs themselves. Their literal complex weights may still cancel, and walks containing H_{erase} edges retain unequal-residual sign information.

Proof. By [proposition 2.6](#), the edge sign is

$$\varepsilon_{\xi_\ell}^{m_{\ell-1}m_\ell} = \mu(d_{m_{\ell-1}})\mu(d_{m_\ell}).$$

Every vertex gauge occurs twice in the closed product, so the product is one. $\square$

The telescoping proposition explains why the common residual/source gauge is not the arithmetic cancellation sought here. It disappears from every closed walk automatically. The signs

$$\prod_{\ell=1}^{2q} \mu(m_{\ell-1}j(\xi_\ell) + h_0)\mu(m_\ell j(\xi_\ell) + h_0)$$

in (42) are the genuine target signs. The proposition above identifies the pure same-residual subfamily on which they are trivial. Bounding the remaining complete weighted sum is an L2 problem.

5.4 Why a support walk count is insufficient

Taking absolute values term by term in (42) reduces the method to a positive walk count and loses its intended advantage. Conversely, a random-sign heuristic for the decorated walks is not a proof for the one literal Möbius array. A valid operator-walk input must estimate the complete fixed- h_0 weighted sum, uniformly over all start rows needed by Ω_{2q} , or the complete closed-walk sum needed by $\text{tr } H^{2q}$.

6 The signed native certificate lattice

The preceding theorems take absolute values at different stages. Their conclusions can be compared only after each is evaluated on the same complete literal matrix.

Definition 6.1 (Available certificates). For the gauged alias operator H , define

$$C_{\text{star}} := \sqrt{\Delta_H} s_*(H), \quad (44)$$

$$C_{\text{pair}} := \rho(P), \quad (45)$$

$$C_{\text{walk},q} := \Omega_{2q}(H)^{1/(2q)}, \quad (46)$$

$$C_{\text{mom},q} := (\text{tr } H^{2q})^{1/(2q)} \quad (q \geq 1). \quad (47)$$

Here P may be the optimal literal pairing-defect matrix or any other proved symmetric nonnegative pair majorant. Put

$$\mathcal{C}_X := \min \left\{ C_{\text{star}}, C_{\text{pair}}, \inf_{q \geq 1} C_{\text{walk},q}, \inf_{q \geq 1} C_{\text{mom},q} \right\}. \quad (48)$$

At a fixed finite X , all terms are finite and computable in principle. The infima over q are mathematical certificates; their exact evaluation by arbitrarily high powers is not claimed to be computationally efficient.

Theorem 6.2 (Exact native eigenvalue certificate). *For the atomic-normalized literal missing-native Gram $K = I + H$,*

$$\kappa_M \leq 1 + \mathcal{C}_X, \quad (49)$$

$$\alpha_M \geq \max\{0, 1 - \mathcal{C}_X\}. \quad (50)$$

Every individual term in (48) may replace $\mathcal{C}_X$.

Proof. Theorems 4.1, 4.4, 5.1 and 5.2 shows that $\|H\|$ is at most every listed certificate. Hence $\|H\| \leq \mathcal{C}_X$. Since

$$\lambda_{\max}(H) \leq \|H\|, \quad \lambda_{\min}(H) \geq -\|H\|,$$

the eigenvalue identities (15)–(16) give the two bounds. Positivity of K supplies the additional zero in (50). $\square$

Remark 6.3 (Exact but not automatically small). The theorem is an exact L0 implication. It does not say that $\mathcal{C}_X = o(1)$, $X^{o(1)}$, or even bounded. Computing a large certificate exactly is diagnostically useful but does not prove cancellation.

6.1 Full-support components

Let the actual collision graph of H have connected components $\mathcal{C}_X$. After a simultaneous permutation of rows,

$$H = \bigoplus_{C \in \mathcal{C}_X} H_C, \quad K = \bigoplus_{C \in \mathcal{C}_X} (I_C + H_C).$$

Proposition 6.4 (Worst-component rule). *One has*

$$\|H\| = \max_{C \in \mathcal{C}_X} \|H_C\|, \tag{51}$$

$$\kappa_M = \max_{C \in \mathcal{C}_X} \lambda_{\max}(I_C + H_C), \tag{52}$$

$$\alpha_M = \min_{C \in \mathcal{C}_X} \lambda_{\min}(I_C + H_C). \tag{53}$$

Thus a uniform theorem must include every actual component. Selecting a favorable component or deleting a difficult row changes the operator.

Proof. These are the standard norm and extremal-eigenvalue formulas for a finite orthogonal direct sum. $\square$

6.2 Quantifier-safe finite certificates

The pairing τ , Abel ordering, Perron weight a , power q , and component decomposition may all be chosen after the finite literal matrix is known. They still yield coefficient-uniform operator inequalities because none depends on the vector to which H is later applied. What is forbidden is to choose or alter the physical carrier, erase extra output labels, delete active rows, or rebuild the certificate separately for each later coefficient vector while advertising one uniform operator bound.

7 The shorted boundary for full reassembly

The preceding sections control the literal missing-native map relative to its own atomic diagonal. Full reassembly introduces a represented map and coefficient directions invisible to its denominator. We now state the exact L0 relation, following the shorted-Gram framework of TPC–60. The underlying finite-dimensional shorted-operator identity is classical [2].

7.1 Direct and shorted Grams

Let $\mathcal{K}, \mathcal{Y}$ be finite-dimensional Hilbert spaces and let

$$A, M : \mathcal{K} \longrightarrow \mathcal{Y}, \quad B := A + M.$$

Assume $A \neq 0$, and define the uniform reassembly constant

$$\delta(A, M) := \inf_{Az \neq 0} \frac{\|Bz\|^2}{\|Az\|^2}. \quad (54)$$

Put

$$N_A := \text{Ker } A, \quad S_A := N_A^\perp, \quad A_{S_A} := A|_{S_A}, \quad M_{S_A} := M|_{S_A}, \quad B_{S_A} := B|_{S_A},$$

and

$$R_A := A_{S_A}^* A_{S_A} > 0 \quad \text{on } S_A. \quad (55)$$

Let

$$W_{\text{nuis}} := B(N_A) = M(N_A) \quad (56)$$

and let $P_{W_{\text{nuis}}}$ be its orthogonal projection. This nuisance space is unrelated to the row mass $W_M(m)$.

Definition 7.1 (Direct Gram and shorting penalty). Define on S_A

$$G_{\text{dir}} := R_A^{-1/2} B_{S_A}^* B_{S_A} R_A^{-1/2}, \quad (57)$$

$$C_N := R_A^{-1/2} B_{S_A}^* P_{W_{\text{nuis}}} B_{S_A} R_A^{-1/2} \succeq 0, \quad (58)$$

$$G_{\text{sh}} := G_{\text{dir}} - C_N. \quad (59)$$

Theorem 7.2 (Exact direct-minus-nuisance formula). *The shorted operator is positive semidefinite and*

$$\boxed{\delta(A, M) = \lambda_{\min}(G_{\text{sh}}).} \quad (60)$$

If

$$G_{\text{dir}} = I + H_{\text{dir}}, \quad \|H_{\text{dir}}\| \leq c, \quad \|C_N\| \leq \eta, \quad (61)$$

then

$$\boxed{\delta(A, M) \geq \max\{0, 1 - c - \eta\}.} \quad (62)$$

If $W_{\text{nuis}} = \{0\}$, then $C_N = 0$ and no shorting penalty occurs.

Proof. For $z = s + n$, with $s \in S_A$ and $n \in N_A$, the denominator is $\|A_{S_A} s\|^2$, while minimization over n gives

$$\inf_{n \in N_A} \|B_{S_A} s + Bn\|^2 = \|(I - P_{W_{\text{nuis}}})B_{S_A} s\|^2.$$

Hence

$$\delta(A, M) = \inf_{s \neq 0} \frac{\langle B_{S_A}^* (I - P_{W_{\text{nuis}}}) B_{S_A} s, s \rangle}{\langle R_A s, s \rangle}.$$

Rayleigh–Ritz after the substitution $x = R_A^{1/2} s$ proves (60) and the positivity of G_{sh} . Under (61),

$$\lambda_{\min}(G_{\text{sh}}) \geq 1 - \|H_{\text{dir}}\| - \|C_N\| \geq 1 - c - \eta.$$

Positivity adds the maximum with zero. If $W_{\text{nuis}} = \{0\}$, its projection vanishes. $\square$

Remark 7.3 (The missing-native H is not H_{dir}). The main operator H in (13) comes from $M^* M$ normalized by the missing atomic diagonal. By contrast, H_{dir} contains

$$A_{S_A}^* M_{S_A} + M_{S_A}^* A_{S_A} + M_{S_A}^* M_{S_A}$$

after represented normalization. No identity in this paper identifies those cross terms with the two-affine formula (18). The missing-native certificate must be combined with a separate mass comparison before it can bound H_{dir} .

7.2 A safe direct-norm corollary

Corollary 7.4 (Canonical perturbation plus nuisance penalty). *Suppose*

$$\|M_{S_A}s\| \leq \varepsilon \|A_{S_A}s\| \quad (s \in S_A). \quad (63)$$

Then

$$\|G_{\text{dir}} - I\| \leq 2\varepsilon + \varepsilon^2. \quad (64)$$

Consequently, if also $\|C_N\| \leq \eta$, then

$$\delta(A, M) \geq \max\{0, 1 - 2\varepsilon - \varepsilon^2 - \eta\}. \quad (65)$$

Proof. The map $U := A_{S_A}R_A^{-1/2}$ is an isometry and $V := M_{S_A}R_A^{-1/2}$ has norm at most ε . Therefore

$$G_{\text{dir}} = (U + V)^*(U + V) = I + U^*V + V^*U + V^*V,$$

which gives (64). Apply [theorem 7.2](#). $\square$

Corollary 7.5 (How the missing-native certificate can enter). *Assume that $\mathcal{K} = \mathcal{K}_X$, that $M = M_X$ is the literal missing-native map, and that its full physical diagonal is D_M from (6). Assume on S_A that*

$$\langle D_M z, z \rangle \leq \chi \|Az\|^2. \quad (66)$$

Then the exact missing-native upper bound gives

$$\|Mz\|^2 \leq \kappa_M \langle D_M z, z \rangle \leq \kappa_M \chi \|Az\|^2.$$

Thus [corollary 7.4](#) applies with

$$\varepsilon = (\kappa_M \chi)^{1/2}. \quad (67)$$

The conclusion still requires a separate bound for C_N .

Proof. The first inequality is the definition of κ_M ; the second is (66). Taking square roots yields (63). $\square$

This corollary is a bookkeeping interface, not a proved physical estimate: neither a favorable χ nor a favorable shorting penalty is obtained in this paper.

8 Five separating countermodels

The examples below are finite-dimensional. Their purpose is not to model the full arithmetic carrier, but to show exactly which inference fails without additional physical input.

Example 8.1 (A residual gauge becomes genuine after erasure). *Let*

$$T : \mathbb{C} \longrightarrow \mathbb{C}^2, \quad Tz = (z, z),$$

and let $E(z_1, z_2) = z_1 + z_2$. The diagonal residual gauge

$$U = \text{diag}(1, -1)$$

is unitary, so T and UT have the same residual norm and the same support magnitudes. Nevertheless,

$$\|ETz\|^2 = 4|z|^2, \quad \|EUTz\|^2 = 0.$$

Thus the location of a residual unit phase is a coordinate choice only before erasure. Once two residual coordinates meet, their relative phase changes the native Gram.

Example 8.2 (Small pair entries can form a large operator). For $r \geq 2$, let

$$H_r := r^{-1/2}(J_r - I_r), \quad K_r := I_r + H_r.$$

Every off-diagonal pair has magnitude $r^{-1/2} \rightarrow 0$, and $K_r \succeq 0$ because the eigenvalues of H_r are

$$\frac{r-1}{\sqrt{r}}, \quad -\frac{1}{\sqrt{r}} \quad (r-1 \text{ times}).$$

However

$$\|H_r\| = \frac{r-1}{\sqrt{r}} \asymp \sqrt{r}.$$

This is the degree-times-pair obstruction in its simplest positive-Gram form.

Example 8.3 (A scalar signed star sum is not an operator bound). On vertices $0, 1, 2, 3, 4$, let H be the Hermitian star with

$$(H_{01}, H_{02}, H_{03}, H_{04}) = (1, -1, i, -i)$$

and $H_{n0} = \overline{H_{0n}}$. Then

$$\sum_{n=1}^4 H_{0n} = 0, \quad s_0(H) = 2, \quad \|H\| = 2.$$

Indeed, a Hermitian weighted star has nonzero eigenvalues $\pm s_0(H)$. For $0 < a \leq 1/2$, the matrix $I + aH$ is positive semidefinite, while the distinguished scalar row sum remains zero and $\|aH\| = 2a$. Cancellation against the all-ones vector therefore says nothing uniform about the star.

Example 8.4 (Support and magnitudes do not determine the norm). Let $r = 2^k$ with $k \geq 2$, let S_r be a symmetric Sylvester Hadamard matrix, and put

$$Q_r := S_r - \text{diag}(S_r).$$

Both $J_r - I_r$ and Q_r have complete off-diagonal support and every off-diagonal entry has magnitude one. But

$$\|J_r - I_r\| = r - 1,$$

whereas, using $S_r^2 = rI_r$,

$$\|Q_r\| \leq \|S_r\| + \|\text{diag}(S_r)\| \leq \sqrt{r} + 1.$$

After multiplying both matrices by $a = (r-1)^{-1}$, they still have identical support and entry magnitudes; both $I + a(J_r - I_r)$ and $I + aQ_r$ are positive semidefinite. Indeed,

$$a\|Q_r\| \leq \frac{\sqrt{r} + 1}{r - 1} \leq 1 \quad (r \geq 4),$$

while the spectrum of $a(J_r - I_r)$ is $\{1, -(r-1)^{-1}, \dots, -(r-1)^{-1}\}$. Also

$$\|Q_r\| \geq \frac{\|Q_r\|_F}{\sqrt{r}} = \sqrt{r-1},$$

because Q_r has $r(r-1)$ unit-modulus off-diagonal entries. Their defect norms differ by a factor of order $\sqrt{r}$. A support graph, codegree, and entry magnitudes cannot recover the missing phase information.

Example 8.5 (Canonical success can be destroyed by shorting). *Let*

$$\mathcal{K} = \mathbb{C}^2, \quad \mathcal{Y} = \mathbb{C},$$

and define

$$A(z_1, z_2) = z_1, \quad M(z_1, z_2) = z_2, \quad B(z_1, z_2) = z_1 + z_2.$$

Then $S_A = \mathbb{C}(1, 0)$, and on S_A the full map agrees exactly with the represented map:

$$G_{\text{dir}} = I.$$

But $N = \mathbb{C}(0, 1)$ and $B(N) = \mathbb{C}$, so

$$C_N = I, \quad G_{\text{sh}} = 0, \quad \delta(A, M) = 0.$$

Equivalently, the nuisance choice $z_2 = -z_1$ annihilates the full output without changing the denominator. A canonical representative estimate cannot replace the shorted quotient.

8.1 Logical separation

[Example 8.1](#) prevents the residual unitary gauge from being called cancellation. [Example 8.2](#) prevents a small pair estimate from being promoted without aggregation. [Example 8.3](#) forces an ℓ^2 -star or stronger uniform object. [Example 8.4](#) separates literal phases from support geometry. [Example 8.5](#) forces the nuisance penalty in full reassembly. These obstructions are independent: repairing one does not repair the others.

9 Conditional arithmetic routes and exponent ledgers

This section states transparent sufficient premises for an asymptotic family $X \rightarrow \infty$. None of the premises below is proved in this paper. Every $o(1)$ is required to be uniform over all active rows, row pairs, blocks, sides, and components in the declared physical scope.

9.1 Route P: uniform literal pair defects

Proposition 9.1 (Conditional pair route). *Assume that the literal pair-defect matrix satisfies*

$$\Delta_P \leq X^{\delta+o(1)}, \tag{68}$$

$$P_{mn} \leq X^{-\tau+o(1)} \quad (m \neq n). \tag{69}$$

Then

$$\|H\| \leq X^{\delta-\tau+o(1)}. \tag{70}$$

Thus this route gives a subpower certificate only when $\tau \geq \delta$, and gives decay only when $\tau > \delta$.

Proof. Apply [corollary 4.6](#) with the two assumed exponents. $\square$

The hypothesis concerns the completed fixed- h_0 pair entries with their literal complex weights. An averaged shift estimate, a density-one set of row pairs, or a reduced-cell support count does not imply (69).

9.2 Route S: uniform star cancellation

Proposition 9.2 (Conditional star route). *Assume (68) and*

$$\max_m \left(\sum_{n \neq m} P_{mn}^2 \right)^{1/2} \leq X^{-\sigma+o(1)}. \quad (71)$$

Then

$$\|H\| \leq X^{\delta/2-\sigma+o(1)}. \quad (72)$$

This route gives a subpower certificate when $\sigma \geq \delta/2$, and decay when $\sigma > \delta/2$.

Proof. Use [corollary 4.3](#). □

An estimate for one scalar sum $\sum_n H_{mn}$ is insufficient by [proposition 4.2](#). The premise must control the entire ℓ^2 -star for every actual row.

9.3 Route W: complete operator walks

Proposition 9.3 (Conditional operator-walk route). *Let $q_X \geq 1$ be integers. If*

$$\Omega_{2q_X}(H) \leq X^{2q_X\nu+o(q_X \log X)}, \quad (73)$$

then

$$\|H\| \leq X^{\nu+o(1)}. \quad (74)$$

The same conclusion follows from the analogous bound

$$\text{tr } H^{2q_X} \leq X^{2q_X\nu+o(q_X \log X)}.$$

Proof. Take $2q_X$ -th roots in [theorems 5.1](#) and [5.2](#). □

When $q_X \rightarrow \infty$, the error term in (73) must be uniform at that growing walk length. A heuristic square-root cancellation for independent signs, a fixed- q estimate with uncontrolled constants, or a count of unsigned support walks is not the stated premise.

9.4 Missing mass, represented scale, and strict reassembly

Suppose the physical represented and missing maps are now placed in the same declared scope. Let $\mathcal{E}_X(z)$ be one fixed reference energy, put $S_A = (\text{Ker } A)^\perp$, and assume uniformly for $z \in S_A$ that

$$\langle D_M z, z \rangle \leq X^{-\sigma_M+o(1)} \mathcal{E}_X(z), \quad (75)$$

$$\|Az\|^2 \geq X^{-\lambda_{\text{rep}}+o(1)} \mathcal{E}_X(z), \quad (76)$$

$$\kappa_M \leq X^{\nu_M+o(1)}. \quad (77)$$

The certificate lattice may be used to prove the last inequality only if one of the preceding L2 routes is actually established.

Proposition 9.4 (Conditional strict exponent gate). *Under (75)–(77),*

$$\varepsilon_X^2 := \sup_{z \in S_A \setminus \{0\}} \frac{\|Mz\|^2}{\|Az\|^2} \leq X^{\nu_M-\sigma_M+\lambda_{\text{rep}}+o(1)}. \quad (78)$$

Hence

$$\boxed{\sigma_M > \nu_M + \lambda_{\text{rep}}} \quad (79)$$

is sufficient for $\varepsilon_X = o(1)$. If, in addition,

$$\|C_N\| = o(1), \tag{80}$$

then

$$\delta(A, M) \geq 1 - o(1).$$

Proof. Combine the three assumed inequalities to obtain (78). The strict inequality (79) makes its right side tend to zero. Then corollary 7.4 and (80) give the final assertion. $\square$

Equality in (79) gives only a subpower upper bound and does not imply that the perturbation tends to zero. Also, no estimate for C_N follows formally from κ_M : the nuisance output $B(N)$ is a separate geometric object.

9.5 Endpoint ledger

The exact L0 identities and the L1 carrier crosswalk have zero exponent cost. Every positive exponent in $\delta, \tau, \sigma, \nu_M, \sigma_M, \lambda_{\text{rep}}$, every block summation, and every comparison between reference energies must be charged once in the common endpoint ledger. If the literal physical family forces total loss at least $1/400$, the endpoint route must stop. Reallocating the same loss among pair, star, walk, or reassembly notation does not repair the budget.

10 Conclusions, kill criteria, and the next decision

10.1 What is proved

For the complete literal missing-native map at one fixed nonzero shift:

- (1) physical atomic normalization gives a positive Gram $K = I + H$ with a Hermitian zero-diagonal alias defect;
- (2) the common source/residual row gauge exposes the exact signs $\mu(mj + h_0)\mu(nj + h_0)$ without changing any spectral quantity;
- (3) on a same-residual row-pair collision this target sign freezes to $\mu(d_m)\mu(d_n)$, so there is no pair-internal Möbius oscillation in H_{res} ; correspondingly, every pure H_{res} closed walk has target-sign product one;
- (4) sign-reversing partial matchings and ordered Abel sums give exact complex-weight pair certificates;
- (5) row stars satisfy the exact sandwich $s_* \leq \|H\| \leq \sqrt{\Delta_H} s_*$, and the local extremizer forbids replacing the star by a distinguished scalar row sum;
- (6) a symmetric pair majorant gives the reducible-safe Perron optimum $\|H\| \leq \rho(P)$;
- (7) even power rows and even traces retain cancellation at the level of complete decorated operator walks, but genuinely new signed information comes only from walks containing H_{erase} edges or from literal complex-weight phases; and
- (8) every certificate bounds both extremal eigenvalues of K , while the full represented-plus-missing problem is exactly a direct Gram minus the positive nuisance penalty C_N .

These are L0 theorems. Their identification with the actual fixed- h_0 missing carrier, complete native output labels, and raw atomic masses is L1.

10.2 What remains unproved

The paper does not prove:

- a power saving for one literal pair defect uniformly over all colliding active row pairs;
- an ℓ^2 -star saving that beats half the actual degree exponent;
- a fixed- or growing-length signed operator-walk estimate with constants uniform in every physical block;
- the required missing-mass saving or represented lower-frame comparison in the same scope;
- a bound $C_N = o(1)$ for the nuisance shorting penalty; or
- any parity estimate, prime-pair theorem, or twin-prime conclusion.

10.3 Kill criteria

The following stops have different strengths.

- (K1) **Wrong object.** If a proposed proof erases γ , side, external block, opposite row, or any label still orthogonal in the native range, or replaces complete outputs by reduced cells, it does not estimate the operator in this paper.
- (K2) **Wrong quantifier.** If the carrier, partition, pairing, or row deletion depends on the later coefficient vector or extremizing eigenvector, it is not a coefficient-uniform certificate.
- (K3) **Wrong physical regime.** A fixed-pair result, average over pairs, average over shifts, density-one row result, or frozen/smoothed family does not imply the required growing-row theorem at the prescribed fixed h_0 .
- (K4) **Pair-route stop.** Failure of $\tau \geq \delta$, or a pair majorant with large $\rho(P)$, kills only the pair/Perron certificate. It does not rule out cancellation between complete stars or operator walks.
- (K5) **Star-route stop.** A star bound that does not beat the target after multiplication by $\sqrt{\Delta_H}$ —equivalently, $\sigma < \delta/2$ in the exponent model of [proposition 9.2](#)—kills only the star certificate. It does not rule out a later even-walk or trace estimate.
- (K6) **Walk-route stop.** An unsigned support-walk count or a nonuniform fixed-moment estimate is not evidence for [\(73\)](#). Failure of one moment length does not kill every other length.
- (K7) **Hard uniform stop.** If the actual shorted Gram G_{sh} has a kernel, equivalently if some z satisfies $Az \neq 0$ and $(A + M)z = 0$, then $\delta(A, M) = 0$. No canonical-representative estimate can repair that uniform reassembly claim. A distinguished physical vector remains a logically separate question.
- (K8) **Endpoint stop.** If the total literal loss is at least $1/400$, the current endpoint implication must be abandoned even if every intermediate qualitative statement is correct.

10.4 Route decision after TPC-68

TPC-67 and the present paper now expose two independent finite-dimensional doors on actual coefficients:

all-row determinant mass and signed native operator walks.

The determinant door needs physical matching, pivot, and cycle-activity input. The signed-operator door needs one of the literal pair, star, or walk premises above, followed by missing-mass and nuisance control. A useful next paper should attack one *specified* fixed- h_0 decorated walk or star family and either prove a uniform saving or produce an actual coefficient counterexample that closes that door. More support-only repackaging would not advance the proof chain.

10.5 Status line

TPC-68 closes a necessary operator interface and identifies the exact location of the genuine two-affine signs after residual gauge removal. It does not break the arithmetic parity barrier. A negative result for one certificate family would still be publishable as a sharp route obstruction, but it must not be presented as evidence for or against the twin-prime conjecture itself.

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